Ansible Automation - Use Cases

A reference of the use cases by which Ansible Automation Platform is commonly implemented, by domain. Each use case is described in the same format (Description, Collection, Key content, Notes) and mapped to Red Hat Certified and Validated content.

5 solution use cases · 7 product use cases · generated 2026-10-21 · community reference, not an official Red Hat publication

Contents

1. AI & Lightspeed
    - Solution use cases (5)
        - Call MCP server tools from playbooks
        - Build execution environments with MCP servers
        - Provision infrastructure for Red Hat AI
        - Configure and serve models on RHEL AI
        - Add AI safeguards to job templates with NeuBird
    - Product use cases
        - Model Context Protocol (MCP) (2)
        - Model Context Protocol (MCP) servers (2)
        - Red Hat AI infrastructure (2)
        - Red Hat Enterprise Linux AI (1)

AI & Lightspeed

AI-assisted automation and automation for AI infrastructure.

Solution use cases

Use Case: Call MCP server tools from playbooks

Description	Model Context Protocol (MCP) servers offer tools for cloud services, code hosting and other systems in one standard way. Playbooks can call those tools as tasks, so MCP-based capabilities run under Automation Platform's credentials, approvals, logging and schedules, and not from an uncontrolled client.
Collection	`ansible.mcp` (Certified), `ansible.mcp_builder` (Validated), `ansible.controller` (Certified)
Key content	• `ansible.mcp.server_info` — Retrieve MCP server information • `ansible.mcp.tools_info` — Retrieve a list of supported tools from an MCP server • `ansible.mcp.run_tool` — Call a specific tool on an MCP server • `ansible.mcp.mcp` — Persistent connection to an Model Context Protocol (MCP) server • `ansible.controller.job_template` — create, update, or destroy job templates.
Notes	• The playbook pattern also lets an Event-Driven Ansible rulebook trigger MCP-based steps, see Build self-healing infrastructure from monitoring alerts. • Treat tool arguments from an AI agent as untrusted input: validate them in the playbook. • Check the Technology Preview support status before committing to production use.

Use Case: Build execution environments with MCP servers

Description	MCP servers need a defined runtime: Python or Node packages, binaries and configuration. Building them into an execution environment gives automation a tested, versioned image that runs the same on a laptop and on the automation controller.
Collection	`ansible.mcp_builder` (Validated), `ansible.mcp` (Certified)
Key content	• `ansible.mcp_builder.install_mcp` — Install and configure MCP Servers • `ansible.mcp_builder.common` — Sets up a generic MCP build environment and installs MCP servers from multiple sources (Go, npm, PyPI) • `ansible.mcp_builder.aws_core_mcp` — Installs the AWS Core MCP server from PyPI with dynamic proxy server strategy • `ansible.mcp_builder.aws_ccapi_mcp` — Installs the AWS Cloud Control MCP server from PyPI • `ansible.mcp_builder.azure_mcp` — Installs the Azure MCP server from npm to interact with Azure resources • `ansible.mcp_builder.github_mcp` — Installs the GitHub MCP server with support for local (build from source) and remote modes

Notes	• Call the servers from playbooks with Call MCP server tools from playbooks. • General guidance on image building is in the platform use case on execution environments. • Pin versions, because MCP servers change quickly.

Use Case: Provision infrastructure for Red Hat AI

Description	GPU-enabled hosts for Red Hat AI are expensive and are often needed for short periods. The playbooks create the instances or bare-metal hosts, install the access proxy and remove everything when the work is done, so teams get an environment on request and do not pay for idle machines.
Collection	`infra.ai` (Validated), `redhat.ai` (Certified)
Key content	• `infra.ai.aws_provision` — Provision infrastructure • `infra.ai.azure_provision` — Provision Azure infrastructure • `infra.ai.gcp_provision` — Create a GCP instance • `infra.ai.baremetal_provision` — Provision baremetal infrastructure • `infra.ai.baremetal_iso` — A role to create ISO image for automated Red Hat AI installation on baremetal host. • `infra.ai.nginx_proxy` — A role to create nginx reverse proxy with self-signed TLS certificates. • `infra.ai.aws_teardown` — Teardown infrastructure
Notes	• Self-signed certificates in the proxy role suit tests. Use certificates from your own CA for shared environments. • Pair with the cloud use cases on tagging and cost hygiene.

Use Case: Configure and serve models on RHEL AI

Description	After the host exists, the AI software has to be configured and the model must be served. The automation initialises InstructLab, sets the configuration, downloads the model, starts the serving process and sends a test request, so every host is set up the same way and a failed host can be rebuilt without notes.
Collection	`redhat.ai` (Certified), `infra.ai` (Validated)
Key content	• `redhat.ai.ilab_init` — Create initial InstructLab configuration • `redhat.ai.ilab_config` — Edit existing InstructLab configuration • `redhat.ai.ilab_model_download` — Download InstructLab AI model • `redhat.ai.ilab_serve` — Serve AI model • `redhat.ai.completion` — Interact with deployed AI model • `redhat.ai.ilab` — Setup RHEL AI configuration
Notes	• Use Build execution environments with MCP servers and Call MCP server tools from playbooks for AI agent integrations. • Limit access to the endpoint through the proxy created during provisioning.

Use Case: Add AI safeguards to job templates with NeuBird

Description	AI-assisted operations only work if automation stays under control. The NeuBird AI gateway for Automation Platform makes job templates aware of the AI service: templates are tagged by safety level, given context before they run, checked before execution and reported on afterwards, with an audit trail.
Collection	`neubird.aap` (Certified), `ansible.controller` (Certified), `ansible.eda` (Certified)
Key content	• `ansible.controller.job_template` — create, update, or destroy job templates. • `ansible.controller.workflow_job_template` — create, update, or destroy Automation Platform Controller workflow job templates. • `ansible.controller.notification_template` — create, update, or destroy Automation Platform Controller notification. • `ansible.eda.rulebook_activation` — Manage rulebook activations in the EDA Controller
Notes	• The collection is certified and published by NeuBird AI. Details on its modules and roles are in its README. • Use this together with Call MCP server tools from playbooks when agents reach automation through MCP. • Start with read-only templates, then add change actions step by step.

Product use cases

Grouped from the modules of each certified or validated collection. Key content lists the components each use case supports, with the modules that manage them (create, update, delete) and query them (_info).

Model Context Protocol (MCP) `ansible.mcp`

Certified Red Hat · version 1.1.0 · updated 2026-05-07 · 3 modules → 2 use cases

Use Case: Call tools on MCP servers

Description	Connect to an MCP server, list its tools and server information, and call a specific tool.
Collection	`ansible.mcp` (Certified, Red Hat)
Key content	**MCP connection** connection `mcp` **Server information** `server_info` **Tool list** `tools_info` **Tool calls** `run_tool`

Use Case: Automate cloud and source control with MCP servers

Description	Example playbooks that use MCP servers to create AWS resources, create an Azure SQL database and create a GitHub repository and pull request.
Collection	`ansible.mcp` (Certified, Red Hat)

Key content	**AWS Cloud Control API demo** playbook `aws_ccapi_demo.demo` **AWS Cloud Control API validation failures** playbook `aws_ccapi_demo_validation_failure.demo` **AWS Core demo** playbook `aws_core_demo.demo` **AWS Core validation failures** playbook `aws_core_demo_validation_failure.demo` **Azure SQL database** playbook `azure_create_sql_database.playbook` **GitHub repository and pull request** playbook `github_create_repository_and_pull_request.playbook`

Model Context Protocol (MCP) servers `ansible.mcp_builder`

Validated Red Hat · version 1.0.3 · updated 2026-01-16 · 0 modules → 2 use cases

Use Case: Install and configure MCP servers

Description	Install MCP servers from PyPI, npm or source with a generic build environment and a playbook that sets them up.
Collection	`ansible.mcp_builder` (Validated, Red Hat)
Key content	**Install MCP servers** playbook `install_mcp` **Generic MCP build environment** role `common` **GitHub MCP server** role `github_mcp`

Use Case: Install cloud provider MCP servers

Description	Install the AWS Cloud Control, CDK, Core and IAM MCP servers and the Azure MCP server.
Collection	`ansible.mcp_builder` (Validated, Red Hat)
Key content	**AWS Cloud Control MCP server** role `aws_ccapi_mcp` **AWS CDK MCP server** role `aws_cdk_mcp` **AWS Core MCP server** role `aws_core_mcp` **AWS IAM MCP server** role `aws_iam_mcp` **Azure MCP server** role `azure_mcp`

Red Hat AI infrastructure `infra.ai`

Validated Red Hat · version 1.2.0 · updated 2025-09-05 · 0 modules → 2 use cases

Use Case: Provision and tear down AI infrastructure in the cloud

Description	Provision and tear down infrastructure on AWS, Azure and Google Cloud.
Collection	`infra.ai` (Validated, Red Hat)

Key content	**Provision AWS infrastructure** playbook `aws_provision` **Tear down AWS infrastructure** playbook `aws_teardown` **Provision Azure infrastructure** playbook `azure_provision` **Tear down Azure infrastructure** playbook `azure_teardown` **Create a GCP instance** playbook `gcp_provision` **Tear down Google Cloud infrastructure** playbook `gcp_teardown`

Use Case: Prepare bare-metal hosts and proxies

Description	Create installation ISO images for bare-metal hosts and provision infrastructure and an nginx reverse proxy with self-signed TLS certificates.
Collection	`infra.ai` (Validated, Red Hat)
Key content	**Bare-metal installation ISO** role `baremetal_iso` **Provision bare-metal infrastructure** playbook `baremetal_provision` **nginx reverse proxy** role `nginx_proxy` **Install nginx proxy** playbook `proxy` **Install nginx proxy on GCP** playbook `gcp_proxy`

Red Hat Enterprise Linux AI `redhat.ai`

Certified Red Hat · version 1.0.0 · updated 2025-04-23 · 4 modules → 1 use cases

Use Case: Set up and serve InstructLab models on RHEL AI

Description	Initialise and edit InstructLab configuration, download and serve models, and query a deployed model.
Collection	`redhat.ai` (Certified, Red Hat)
Key content	**Set up RHEL AI** playbook `ilab` **Initial InstructLab configuration** `ilab_init` **InstructLab configuration** `ilab_config` **Model download** `ilab_model_download` **Serve a model** role `ilab_serve` **Model completion** `completion`

Other certified and validated collections

Module-level content for these collections could not be read from a public source repository. Browse them on Automation Hub.

`automatics_ai.automatics`	Ansible Collection to interact with the automatics software (Certified, automatics.AI)
`neubird.aap`	Intelligence gateway between Ansible Automation Platform and NeuBird AI. Makes any AAP job template AI-aware with reporting, safety tagging, context injection, pre-flight gating, and audit logging. (Certified, NeuBird AI)